MILITARY SPECIFICATION

DETONATOR, FLASH, PA504 FOR MINE, ANTIPERSONNEL, M16A2

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, loading, assembling and packing for one type of flash detonator designated as Detonator, Flash, PA504 for Mine, Antipersonnel, M16A2.

2. APPLICABLE

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions General Specification For
MIL-P-48250 - Primer, Stab M55 Parts For and Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-650 - Explosives, Sampling, Inspection and Testing
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1345

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

DRAWINGS

US ARMY PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

8796342 - Detonator, Flash PA504, Assembly
9258182 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

INSPECTION EQUIPMENT

8816367 - Firing Pin Guide
8827934 - Adapter, Ball, Primer Sensitivity Testing Fixture
8855292 - Fixture, Functioning Test
8861368 - Disc
9200602 - Bushing
8867811 - Holder, Primer
9200012 - Pin, Firing

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator. - The detonator shall comply with all requirements specified on Drawing (dwg.) 8796342 and with all requirements specified in applicable specifications.

3.3 Moisture content.

3.3.1 Lead azide. - The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent.

3.3.2 Lead styphnate. - The moisture content of the lead styphnate used in loading the detonator shall not exceed 0.30 percent.

3.4 Functioning. - The detonator shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 3/16 inch.

3.5 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.6 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the contracting officer. The first article sample shall consist of twenty-five (25) detonator cups and 315 loaded detonator assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Detonator, Flash, PA504, For Mine Antipersonnel, M16A2 Parts and Loaded Assemblies	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Detonator cups Dwg. 8796341 Examination for defects	25		3.2	4.4.2.1
	Loaded Detonator assemblies Dwg. 8796342 Examination for defects	25(a)		3.2	4.4.2.2
	Functioning test	315(b)		3.4	4.4.3.3
NOTES:	(a) Randomly select 25 from 315 (b). After inspection return to sample (b) for functioning test.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts of one lot interfix from one manufacturer.

b. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. Lead styphnate of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same conditions as specified in "b" above.

4.4.2 Examination. - (See MIL-A-48078) - Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8796341
4.4.2.1	Cup, Detonator				NEXT HIGHER ASSEMBLY 8796342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100%	3.2	Visual
Major B					
131	Inside diameter		0.40%	3.2	Gage
132	Diameter of flange		0.40%	3.2	Gage
133	Thickness through bottom		0.40%	3.2	Gage
134	Length		0.40%	3.2	Gage
135	Bottom not flat		0.40%	3.2	Straight edge
Minor					
201	Small outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8796342
4.4.2.2	Detonator Assembly				NEXT HIGHER ASSEMBLY 8796342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH - REFERENCE /INSPECTION METHOD
Critical					
1	Cup, cracked or split		100%	3.2	Visual
Major B					
131	Length under flange, max		0.40%	3.2	Gage
132	Outside diameter, max		0.40%	3.2	Gage
133	Disc loose, puncture, torn or wrinkled so that charge is exposed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Box, Set-up, Inner, After Loading and Prior to Closing Slide Cover				9258182
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100%	3.2	Visual
2	Filler and cushion assembly improperly assembled		100%	3.2	Visual
3	Stiffner missing		100%	3.2	Visual
4	Slide cover assembly reversed		100%	3.2	Visual
Major B					
131	Release paper missing		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil/ -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Box, Set-up, Outer, with Inner Set-up Boxes Assembled, and Prior to Assembling Top	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up boxes		100%	3.2	Visual
2	Stiffner not facing one direction		100%	3.2	Visual
Major A					
101	Inner box does not fit properly in outer box		0.40%	3.2	Visual
102	Fillers missing		0.40%	3.2	Visual
Major B					
131	Label not securely applied to box		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Box improperly taped		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Sealed Packed Barrier Bag, Prior to Packing in Box				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.2	Visual
102	Evidence of crushed box		0.40%	3.2	Visual
Note:	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Box, Packing, Ammunition, Prior to Assembling Cover and Strapping				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Bag containing set-up boxes damaged		0.40%	3.2	Visual
132	Cushioning material incomplete or missing from outer set-up boxes		0.40%	3.2	Visual
133	Marking missing, misleading or unidentifiable		0.40%	3.2	Visual
134	Bottom cushioning missing or of incorrect thickness		0.40%	3.2	Visual
Minor					
201	Contents loose within box		0.65%	3.2	Visual-Manual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48888 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Box, Packing, Ammunition		9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Strapping missing, broken, loose or improperly fastened		0.40%	3.2	Manual-Visual
102	Box damaged		0.40%	3.2	Visual
103	Marking missing, incomplete or illegible		0.40%	3.2	Visual
Minor					
201	Wooden box cover nailed to box		0.65%	3.2	Visual
202	Metallic seal missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. - (See Note)

NOTE: Chemical sampling. - A representative sample shall be randomly selected for the chemical tests specified in 4.5. The procedure described in ASTM E300 should be used to withdraw the samples. The chemical tests in 4.5 shall be performed using prescribed analytical procedure for replicate determinations given in standard analytical textbooks.

4.4.3.1 Moisture content of lead azide - (see 3.3.1) Major A defect. - The moisture content of the lead azide shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. The sample shall be selected just prior to loading and the moisture content determined as specified in 4.5.1. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the lead azide shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.2 Moisture content of normal lead styphnate - (see 3.3.2) Major A defect. - The moisture content of the normal lead styphnate shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. The sample shall be selected just prior to loading and the moisture content determined as specified in 4.5.2. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the normal lead styphnate shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.3 Functioning - (see 3.4) Major A defect. - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If one or two defects are found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 4 or more defects are found, the lot shall be rejected. The test shall be performed as specified in 4.5.3 using test equipment as specified in 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.4.3.4 Heat seal test of sealed bag - (see dwg. 9258183) Major A defect. - Sampling, acceptance criteria, and test procedures for this test shall be as specified in MIL-P-116.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with 4.4.2.1 and the test methods of 4.5. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.2 herein.

4.5 Test Methods and Procedures.

4.5.1 Moisture content of lead azide. - The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene when weighing.

4.5.2 Moisture content of normal lead styphnate. - Transfer, quickly, approximately two grams of normal lead styphnate to an uncovered tared pyrex dish, approximately 55mm in diameter and 30mm in height. Weigh the sample as quickly as possible to 0.0001 gram and place in an oven maintained at 70°±2°C and dry to constant weight. At the end of the drying period, remove the sample from the oven and immediately place it in a desiccator containing anhydrous calcium chloride. Allow the sample to cool in the desiccator for approximately 20 minutes and then, as quickly as possible, weigh to the nearest 0.0001 gram. Calculate the percentage of moisture in the sample from the loss in weight.

4.5.3 Functioning. - The detonator shall be assembled to the test fixture so that the output end of the M55 primer (see MIL-P-48250) is facing the disc covered end of the detonator. A steel ball 1 3/16 inches in diameter and weighing 3.95 ounces, maximum, shall be dropped on the firing pin to function the M55 primer (which has been inserted in primer holder dwg. 8867811) from a height of 8 inches maximum. The height of drop shall be measured as height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire, or fails to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

create a hole in the lead disc of the required size shall be classed defective. Method of inspection shall be in accordance with dwg. 8816367, 8827934, 8855292, 8816368, 8867811, 9200012 and 9200602.

4.5.4 Heat seal of sealed bag. - The adequacy of the seal on the bag shall be determined in accordance with MIL-P-116.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. - Preservation and packaging shall be as specified on dwgs. 9258182 and 9258183.

5.1.2 Level C. - The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing.

5.2.1 Level C. - Packing shall be as specified on dwg. 9258184.

5.2.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

5.3 Marking. - Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183, and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.2 Submission of Inspection Equipment Designs for Approval (See MIL-A-48078). - Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T-T Dover, New Jersey 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

6.4.1 Functioning. - A random sample of 315 detonators shall be selected from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If 1 or 2 defectives are found, a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample, a total of 5 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.3 using test equipment as specified in 4.4.4.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-157

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.